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Smart city-based e-commerce security technology with improvement of SET network protocol

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ABSTRACT

In this paper, we proposed an improvement of SET protocol for E-commerce transaction. We study the security technology and protocol flow of set protocol, and point out the shortcomings of SET protocol. Due to the lack of security and low scalability of the current protocol, this paper proposes a server e-wallet strategy, which divides the original client into e-wallet client and e-wallet server, and enhances the convenience and security of the transaction by transferring the client to the issuer or the third-party financial institution, while the scalability is stronger. In order to ensure the atomicity of goods and confirm the atomicity of delivery, this paper proposes a secure e-commerce payment protocol based on the fourth party, which includes master protocol, customer return protocol, customer exchange protocol and customer merchant transaction dispute resolution protocol. According to the efficiency of SET protocol, a security hierarchical control model is proposed, which can select different security levels to operate according to the actual needs of customers, and the principle and implementation of the model are analyzed.

1. Introduction

With the rapid development of Internet, computer and network have become increasingly important in people's daily life [1,2]. The businesses carried out by computer and network haves become increasingly diverse, one of which is E-commerce. E-commerce is based on the implementation of the open Internet. Because of its high efficiency, low cost advantages has brought new opportunities for the development of enterprises and gradually become a new business model, it will become the main form business activities of the information society in the future [3–5]. On the Internet, every user is faced with various threats of insecurity, and E-commerce also involves a lot of private personal information, hence safety assurance is particularly important [6].

However, due to its basis on an open and free Internet, the weak security of E-commerce has become a serious problem, which affects greatly the development of E-commerce. Accordingly, in order to develop the E-commerce, it is greatly demanding to solve the security problem, and guarantee the security, integrality, availability, authenticity and non-repudiation of E-commerce [7–10]. Electronic payment is the most core and the most essential link of electronic commerce, is the foundation condition which electronic commerce can carry on smoothly [11–13]. Because the electronic payment needs to carry on in the open network, therefore sensitive information in payment process needs some mechanism to guarantee its security and electronic payment protocol is one of the key technologies to ensure its security [14–16]. Secure electronic transaction SET protocol is proposed by the United States Visa and MasterCard. It is designed to solve the purpose of security transactions between users, businesses and banks, to ensure confidentiality and integrity of information to ensure the identity of the customer and business certification [17–20].

With the proposal of set protocol, it is considered to be the most promising payment protocol in the field of e-commerce, and also a widely used payment protocol [21,22]. Although the security of the protocol is very high, some scholars have found that there are still some shortcomings in the protocol itself, such as the low efficiency of the algorithm of the protocol, the need for repeated identity authentication of all parties to the transaction. Zhang Ruoyan found that the agreement is for western credit cards, while most people in China use debit cards, so it cannot meet the domestic market demand well [23,24]. Wen Tingting found that the agreement could not guarantee the atomicity of the goods that must be met by online payment [25–27]. Therefore, the researchers put forward many improvement schemes for the loopholes of the protocol, for example, Zhang lengti proposed to replace RSA algorithm with ECC algorithm of elliptic curve and DES algorithm with AES algorithm with faster speed and higher security [28]. As the current agreement cannot meet the debit card, Longtao mainly studies the international debit card organization expansion specification, and proposes the expanded specification to meet the debit card payment process [29]. About the atomicity of the agreement, Ren Lili [30] has ensured the cash on delivery principle by increasing the payment gateway as a trusted third party to supervise the data flow of the

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parties to the transaction. With regard to the complexity of the protocol transaction process, Huang Shaoyin, Chen Qingfeng and Sanchez, etc. simplified the protocol identity authentication mode by using one-way identity authentication mode instead of cross identity authentication mode, thus improving the efficiency.

Therefore, improving the SET protocol is of great significance for solving the e-commerce security problem and further promoting the development of e-commerce.

There are five parts in this paper. The first part mainly introduces the background and significance of this paper. The second part puts forward the server electronic wallet scheme model. The third part mainly improves the atomicity of the SET protocol. The fourth part mainly proposes the security hierarchical control model of SET. The fifth part mainly summarizes the whole paper.

2. Model of server e-wallet scheme

2.1. Design background

Because of the complexity of set transaction process, consumers must install a kind of client software called "fat" electronic wallet, which contains the personal certificate, private key and historical payment information of the cardholder. However, every time consumers purchase their own purchase information records are recorded on the personal computer along with the software, which is easy to cause information to be used by hackers. At the same time, when users change computers, they must download and install the client software on the computer again in order to shop, which causes a lot of inconvenience to cardholders. In order to solve this problem, the concept of server wallet is proposed, and the model is shown in Fig. 1.

2.2. Working principle

The server e-wallet is divided into two parts: e-wallet client and e-wallet server.

① electronic wallet client

The client of e-wallet can be realized by installing the corresponding browser plug-in. It is a graphical user interface for the client to manage their account, process payment and view historical transaction record information. In the communication process, it is used to receive the input instruction from the customer and send it to the E-wallet server, and send the processing result to the customer after the server processing. In order to ensure the security of information transmission, the wallet client and server are guaranteed by SSL protocol. There are five layers of client structure, each of which calls the corresponding API to provide services to the upper layer. For example, the encryption algorithm management layer needs to call the encryption and decryption algorithm to realize the encryption and decryption module, and the data transmission layer needs to call open SSL to realize the SSL protocol module.

② wallet server

E-wallet server is usually stored in the issuing bank or other trusted financial institutions. It is responsible for managing many cardholder account information, processing customer payment information and saving customer transaction records. Its functions mainly include: managing the account and historical transaction information of the cardholder; managing the certificate and private key of the cardholder; processing the message transmission information of the server and other transaction entities; receiving the payment request of the electronic wallet client and returning the processing result to the client. The logical structure is shown in Fig. 2.

The client request is sent to the electronic wallet server management layer through the server engine for processing, and the message transmission uses the SSL secure channel to communicate. It is the responsibility of the management to perform the corresponding processing results in accordance with the client's request instructions. When a SET transaction is required, the cardholder and the merchant communicate with each other through the SET transaction component interface to complete the payment request and response. Management is an important part of the entire server. The SET security layer mainly performs operations such as encryption and decryption, log recording, and digital signature of SET messages. The encryption and decryption algorithm module can provide different encryption and decryption algorithms for the server. The HTTP communication module encapsulates the SET communication message of the client and the merchant according to the HTTP protocol format requirement.

3. Improve the atomicity of the SET protocol

As the core of e-commerce, electronic payment has always been a research hotspot, and its atomicity research is even more popular. In 1996, JD of carnegic Mclon university first elaborated the atomic definition of electronic payment, and proposed an electronic payment protocol — netbill, which meets the atomic definition. Unfortunately, this agreement can only be used for payment of digital goods, not for payment of traditional general goods. In fact, the set protocol cannot guarantee the atomicity of goods and the atomicity of full acknowledgment delivery, that is to say, the protocol cannot guarantee that the customer can receive the goods after payment or the goods received are consistent with the order. Moreover, it is difficult to extract transaction evidence and lack of a reasonable dispute handling rule. Therefore, it is necessary to study a set of e-commerce payment protocol to ensure the atomicity of goods and confirm the atomicity of sending.

The participants of secure e-commerce payment protocol based on the fourth party include customers, merchants, Payment institutions and the fourth party. The fourth party is the logistics party, such as postal and other logistics companies, whose responsibilities are: responsible for the delivery of goods, assist payment agencies to solve trade disputes, etc. The secure e-commerce payment protocol based on the fourth party, like set protocol, is mainly applied to B2C operation mode. In this agreement, based on the main agreement of e-commerce payment agreement and considering the possible disputes in the transaction, three kinds of dispute settlement agreements are designed: the return agreement of customer's request for return; the exchange agreement of customer's request for exchange; the dispute settlement agreement of the inconsistency between the goods received by customer and the order of customer.

① main agreement description of payment agreement

In this agreement, the participant is customer C, the merchant is m, the payment institution is p and the fourth party is w. the main agreement process of the payment agreement is shown in Fig. 3. In order to facilitate the formal description of the protocol, the symbols used in the protocol are described, as listed in Table 1.

$$C \rightarrow M: \{(OI' \parallel CerC)\}[OI' = D_{SCK}(OI)] \tag{1}$$

Customer C browses the goods sold by merchant M, agrees to the merchant's quotation, and digitally signs the order OI : $OI' = D_{SKC}(OI)$. Merge OI with customer certificate message: $OI' \parallel CerC$, send the merged message to merchant M.

$$M \rightarrow C: \{Y\}[E_{PKC}(OI') = OI = D_{SKM}(OI'), OI''' = D_{PKW}(OI'')] \tag{2}$$

$$Y = D_{SKM}(Num \parallel CerP \parallel CerM \parallel CerW \parallel Pay \parallel OI''')$$

After receiving the OI': $E_{PKC}(OI') = OI$, the merchant M decrypts the order, confirms that the order is issued by the customer C, and generates a payment request Pay according to the order OI. The dual digital signature OI'' of the merchant M digital signature generation order is encrypted with the W public key: $OI''' = D_{PKW}(OI'')$. The merchant M combines the merchant number with the customer number, time stamp, and random number to generate a transaction unique identifier Num. The merchant M combines *Num, CerP, CerM,*

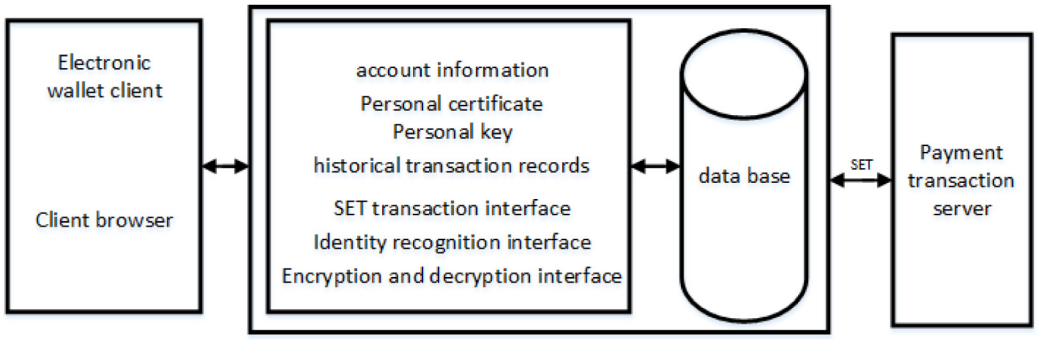


Fig. 1. Server electronic wallet model.

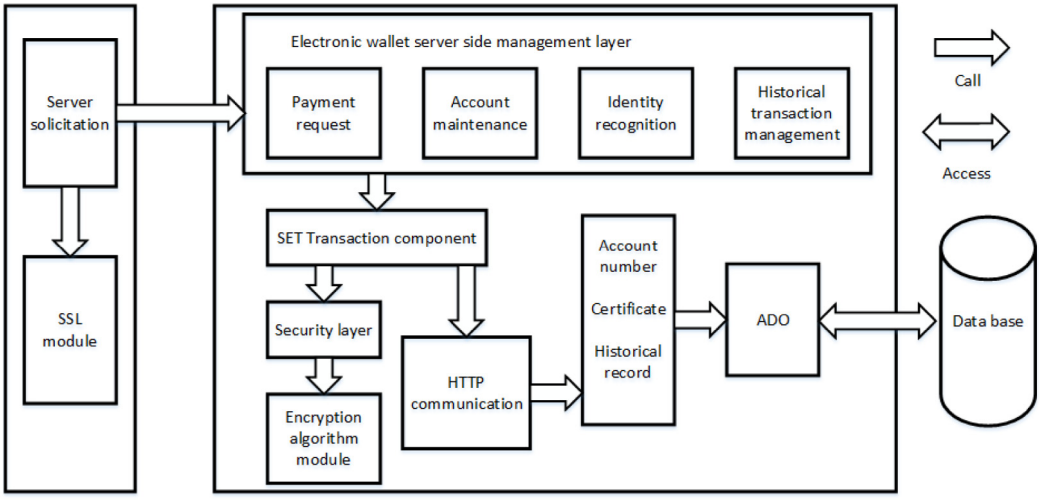


Fig. 2. Electronic wallet server logical structure.

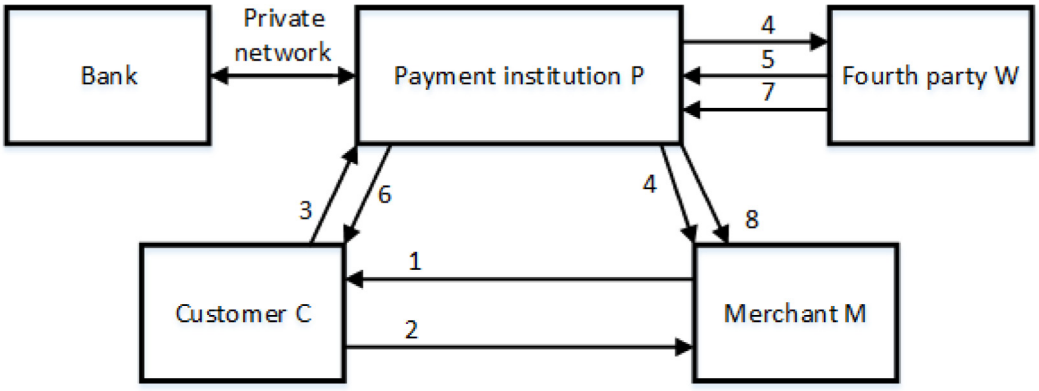


Fig. 3. Master protocol flowchart.

$CerW$ and pay, and OI''' to sign and obtain Y, and then sends Y to the client C.

$$\begin{aligned} C \to P:\ & \{Y_1\}[E_{PKM}(Y) = (Num \parallel CerP \parallel \\ & CerM \parallel CerW \parallel Pay \parallel OI'''), CK(CerP), \\ & PI'' = D_{PKP}(PI'), Y_1 D_{SKC}(Num \parallel CerP \parallel \\ & CerM \parallel CerW \parallel OI''' \parallel PI'')] \end{aligned} \tag{3}$$

Client C decrypts the received Y and the confirmation message is sent by merchant M. Customer C authenticates the certificate: $CK(CerP)$ and proves that P's identity is true and valid.

$$\begin{aligned} P \to M:\ & \{Y_2\}P \to W:\ \{Y_2\}[E_{PKC}(Y_1) = (Num \parallel CerC \parallel CerM \parallel \\ & CerW \parallel OI''' \parallel PI''), \\ & CK(CerC), CK(CerM), CK(CerW), E_{SKP}(PI'') \\ & = PI'', E_{PKC}(PI') = (PI \parallel PAN), \\ & Y_2 = D_{SKP}(Num \parallel T_1 \parallel Msg)]. \end{aligned} \tag{4}$$

Payment gateway P decrypts the received Y_1, and the confirmation is sent by customer C. The payment gateway P verifies the certificate of customer C to verify the identity of customer C; the certificate of merchant m to verify the identity of merchant M; the digital certificate

Table 1
The symbol description table of the Protocol.

Symbol	Explain	Symbol	Explain
C	Customer	$Hmsg_1$	Message of successful exchange application
W	Fourth party	$Qmsg_1$	Fraud handling result message
$\rightarrow$	Message sending	T_3	Customer return delay
Pay	Merchant payment request	$D_{SKC}()$	Customer digital signature
$CerC$	Customer digital certificate	$D_{SKW}()$	Fourth party digital signature
$CerW$	Fourth party digital certificate	$E_{PKP}()$	Verify payment agency digital signature
PKP	Public key of payment institution	P	Payment institution
SKM	Merchant private key	$\|\|$	Message merging
Msg	Payment completion message	PAN	Customer personal account message
$Hmsg$	Request exchange message	PI	Customer payment instruction
$Tmsg$	Request return message	$CerP$	Payment agency digital certificate
$Qmsg$	Fraud statement message	PKM	Merchant private key
T_2	Transaction declaration delay	SKC	Customer private key
$CK()$	Validation of digital certificate	SKW	Fourth party private key
$D_{PKM}()$	Payment agency digital signature	Msg_2	Customer signed in message
$E_{PKM}()$	Verify merchant digital signature	$Tmsg_2$	Returned message
M	Merchant	$Hmsg_2$	Return message of customer in case of exchange
$\{\}$	Messages sent	T_1	Merchant delivery delay
$[]$	Operations performed by the sender	T_4	Return delay in case of customer applying for replacement
OI	Order	$D_{SKM}()$	Merchant digital signature
$CerM$	Merchant digital certificate	$E_{PKC}()$	Verify customer digital signature
PKC	Customer public key	$E_{SKW}()$	Verify fourth party digital signature
PKW	Fourth party public key	Msg_1	Merchant delivered information
SKP	Payment gateway private key	$Tmsg_1$	Successful return message

of the fourth party W to verify the identity of the fourth party W. After the PI "received is decrypted by payment institution $P(PI|||PAN)$, it is confirmed that it is the payment instruction sent by customer C, and the account information pan and payment instruction PI are obtained.

The payment gateway P identifies the payment instruction and checks whether the customer C account has sufficient amount through special channels. If it is insufficient, cancel the transaction; if the account amount of customer C is sufficient, payment gateway P will temporarily print the payment item of customer to a temporary vendor number, and then get the message Msg of successful payment completion. Payment gateway P will sign transaction number Num, merchant supply delay time T_1 and Msg to get Y_2 and send them to m and WW respectively.

$$W \rightarrow P:\ \{Y_3\}[E_{PKP}(Y_2) = (Num \parallel T_1 \parallel Msg),\ Y_3 = D_{SKW}(Num \parallel Msg_1)] \tag{5}$$

The fourth party W decrypted and received Y_2, confirming that it was sent by P. The transaction number Num is added to the transaction database and at the same time, the merchant supply delay time T_1 is turned on, waiting for M to send the product. W receives the product sent by M, and checks whether the delivery delay is within the prescribed time T_1. If the merchant delivers the product on time, he gets the message that the merchant has delivered the product Msg_1 on time. The fourth party W merges the transaction numbers Num and Msg_1 and signs them to Y_3 and sends them to P.

$$P \rightarrow C:\ \{(Num \parallel Msg_1)\}[E_{PKW}\{Y_3\} = (Num \parallel Msg_1)] \tag{6}$$

P decrypts Y_3, confirms that it was sent by W, and sends the transaction number Num and $Msg_1$to the customer, so that customer C can know the transaction situation.

$$W \rightarrow P:\ \{Y_4\}[Y_4 = D_{SKW}(Num \parallel T_2 \parallel Msg_2)] \tag{7}$$

After the fourth party W ships, customer C receives the goods. At the same time, W signs the transaction number Num, the representation delay time T_2, and the product receipt message Msg_2 to get Y_4 to P.

$$P \rightarrow M:\ \{(Num \parallel Msg_2)\}[E_{PKW}\{Y_4\} = (Num \parallel T_2 \parallel Msg_2)] \tag{8}$$

Payment institution P decrypts the received Y_4 and confirms that it was sent by W. At the same time, the delay of appealing the transaction dispute is T_2, and the customer should appeal to the payment gateway within the delay time T_2. At the same time, the transaction number Num and the customer's received product message Msg_2 are sent to M, so that the merchant can know the progress of the transaction.

② Customer return process agreement

If the customer requests a return due to some non-human reason, the customer must appeal to the payment gateway P within the delay time T_2 and start the return agreement. The flow chart of the return process is shown in Fig. 4.

The return process is as follows:

$$C \rightarrow P:\ \{Y_5\}[Y_5 = D_{SKC}(Tmsg \parallel Num)] \tag{9}$$

Customer C wants to apply for a return, and combines transaction number NUM and $Tmsg$ to get Y_5 and sends it to P.

$$P \rightarrow C:\ \{Y_6\}P \rightarrow W\{Y_6\}[E_{PKC}\{Y_5\} = (Tmsg \parallel Num),\ \{Y_6\} = D_{SKP}(Num \parallel T_3 \parallel Tmsg_1)] \tag{10}$$

P decrypted the received Y_5 and confirmed that it was sent by the customer. And check whether the transaction meets the requirements of the return application; if it meets the requirements, the payment institution signs NUM, the delay time T_3 for the return, and the successful return message $Tmsg_1$ to obtain Y_6, and sends Y_6 to C and W respectively.

$$W \rightarrow P:\ \{Y_7\}[E_{PKP}\{Y_6\} = (Num \parallel T_3 \parallel Tmsg_1),\ \{Y_7\} = D_{SKW}(Num \parallel Tmsg_2)] \tag{11}$$

W decrypts the received Y_6, confirms that the message was sent by P, and starts the return delay T_3. The customer applies for the return from transaction number Num to W and prints the return application form. The fourth party checks the delay T_3 and compares the goods for any artificial quality damage. If everything is OK, send the returned goods to the merchant. At the same time, W combines the numbers NUM and $Tmsg_2$, signs them to generate Y_7, and sends them to P.

$$P \rightarrow M:\ \{(Num|||Tmsg_2), E_{PKW}\{Y_7\} = (Num \parallel Tmsg_2)\} \tag{12}$$

P decrypted the received Y_7 and confirmed that it was from W. At the same time, P sends the numbers Num and $Tmsg_2$ to M, so that the merchant knows the transaction process. Finally, P will pay the customer's payment from the temporary account back to the customer's account, cancel the payment, and complete the return process.

③ Customer exchange process agreement

If the customer requests a replacement for some reason, he must apply for a replacement within the replacement delay and start the replacement agreement. The exchange process is shown in Fig. 5.

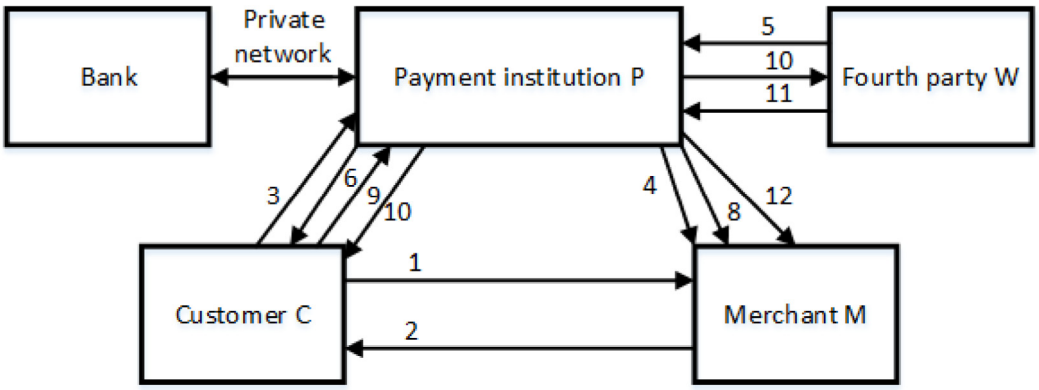


Fig. 4. Flow chart of return process.

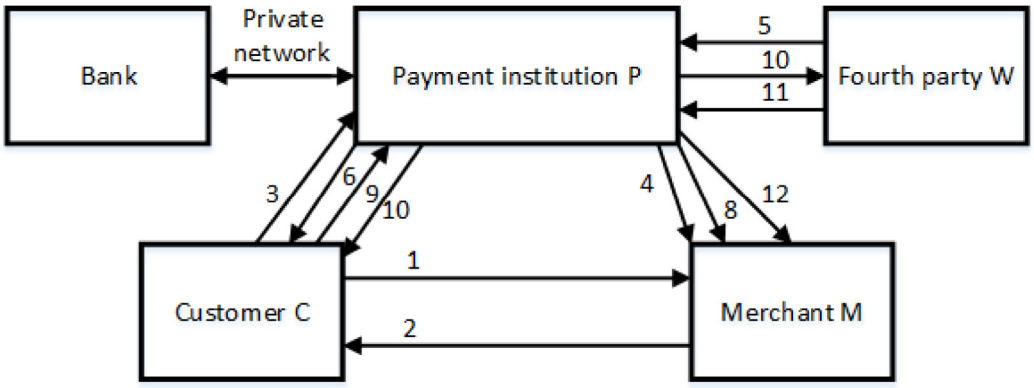


Fig. 5. Replacement process of the protocol.

The exchange process is as follows:

$$C \rightarrow P: \{Y_8\}[Y_8 = D_{SKC}(Hmsg \parallel Num)] \tag{13}$$

The customer merges Num and $Hmsg$, signs it to get Y_8, and sends Y_8 to P.

$$\begin{aligned} &P \rightarrow C: \{Y_9\} P \rightarrow W\{Y_9\}[E_{PKC}\{Y_8\} = (Hmsg \parallel Num),\\ &\{Y_9\} = D_{SKP}(Num \parallel T_4 \parallel Hmsg_1)] \end{aligned} \tag{14}$$

P decrypts the received Y_8, and the authentication is sent by the client. Determine whether the transaction meets the exchange requirements. If the requirements are not met, cancel the replacement application; if the replacements requirements are met agree to the exchange and merge Num and customer C's exchange delay T_4 with $Hmsg_1$ to sign Y_9, and send Y_9 to C and W respectively.

$$\begin{aligned} &W \rightarrow P: \{Y_{10}\}[E_{PKP}\{Y_9\} = (Num \parallel T_4 \parallel Hmsg_1),\\ &\{Y_{10}\} = D_{SKW}(Num \parallel Hmsg_2)] \end{aligned} \tag{15}$$

W decrypts the received Y_9, and the authentication is a message from P and initiates the exchange delay T_4 when returning goods. After receiving the returned merchandise from the customer, W checks whether the merchandise has been damaged by artificial quality. If there is no damage, the Num and the customer's exchanged and returned information $Hmsg_2$ are combined to sign Y_{10} and send it to P.

$$P \rightarrow M: \{(Num||Hmsg_2), E_{PKW}\{Y_{10}\} = (Num \parallel Hmsg_2)\} \tag{16}$$

P decrypts the received Y_{10} and confirms that it was sent by W. P combines Num and $Hmsg_2$ and sends it to M.

The fourth party W, after the merchant M receives the customer's return, requires that M must ship to W within the time limit T_1. At this time, the process shifts to the fifth step of the main agreement of the payment agreement.

④ Customer and merchant dispute process agreement

If the goods received by the customer are not originally purchased, the customer can apply to the payment gateway to submit a dispute. The fraud dispute handling agreement process is shown in Fig. 6.

The formal description of the transaction fraud dispute settlement agreement is as follows:

$$C \rightarrow P: \{Y_{11}\}[Y_{11} = D_{SKC}(Num \parallel Qmsg)] \tag{17}$$

Customer C digitally signs the identification number Num and the transaction fraud appeal message $Qmsg$ to obtain Y_{11}, and sends it to the payment institution P.

$$\begin{aligned} &P \rightarrow W: \{Y_{12}\}[E_{PKC}\{Y_{11}\} = (NuM \parallel Qmsg),\\ &\{Y_{12}\} = D_{SKP}(Num \parallel CerC \parallel CerM \parallel OI''')] \end{aligned} \tag{18}$$

Payment institution P decrypts the received message Y_{11} and confirms that the message was sent by customer C. The transaction identification number Num, the digital certificate $CerC$ of the customer C, the digital certificate $CerM$ of the merchant M, and the double-signed OI''' message of the order information are combined to sign Y_{12}, and sent to the fourth party W.

$$\begin{aligned} W \rightarrow P: &\{Y_{13}\}[E_{PKP}(Y_{12}) = (Num \parallel CerC \parallel\\ &CerM \parallel OI'''), E_{SKW}(OI''') = OI'',\\ &E_{PKW}(OI'') = OI', E_{PKC}(OI') = OI, Y_{13} = D_{SKW}(Num \parallel Qmsg_1)] \end{aligned} \tag{19}$$

The fourth party W receives Y_{12} sent by the payment institution P, decrypts it, and confirms that it was issued by the payment institution P. W decrypts the OI''' multiple times and obtains the order OI, confirming that the message was jointly issued by customer C and merchant M. The fourth party W compares the delivery note of the

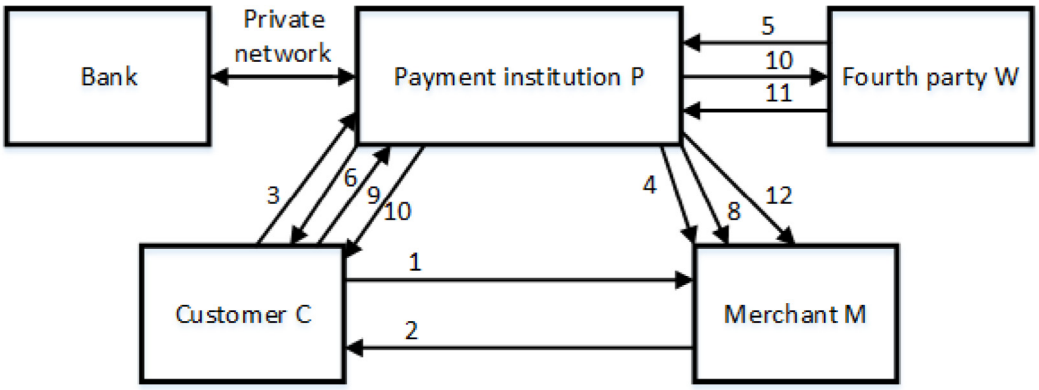


Fig. 6. Transaction processing fraud dispute of the protocol.

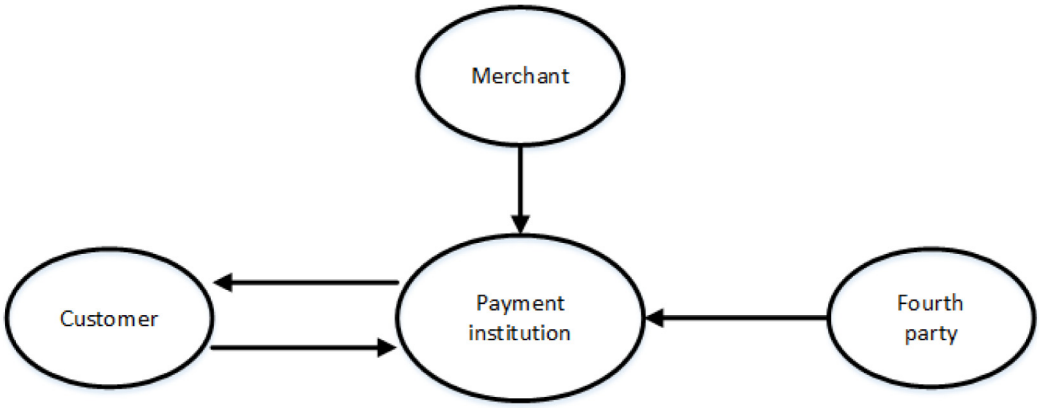


Fig. 7. Authentication model of the protocol.

merchant M stored on the local server with the order information OI, and generates a fraud complaint processing message $Qmsg_1$; if they are the same, it means that the customer C is a malicious appeal; if they are not the same, it means that the merchant M is cheating consumer. Digitally sign Num and fraud processing message $Qmsg_1$ to get Y_{13}, and send Y_{13} to P.

$$P \rightarrow C: \{(Num \parallel Qmsg_1)\}[E_{PKW}\{Y_{13}\} = (Num \parallel Qmsg_1)]or \tag{20}$$

$$P \rightarrow M: \{(Num \parallel Qmsg_1)\}[E_{PKC}\{Y_{13}\} = (Num \parallel Qmsg_1)]$$

P decrypts the received Y_{13}, confirms that it is a message from W, and obtains Num, $Qmsg_1$.

If the customer tells the merchant, send $(Num \parallel Qmsg_1)$ to C and warn the customer C and compensate the merchant for the loss, and end the entire transaction.

If the merchant deceives the consumer, send $(Num \parallel Qmsg_1)$ to M and warn the merchant M and compensate the customer for the loss. At this time, the agreement turns to the tenth step of the return agreement. Continue through the entire return process.

3.1. Protocol security analysis and comparison

The purpose of this agreement is to solve the atomicity of goods and confirm the atomicity of sending in electronic payment protocols, in order to ensure the security of online payments and avoid the occurrence of fraud in e-commerce transactions.

The parties involved in the agreement are customers, merchants, payment institutions and fourth parties. The authentication mode is shown in Fig. 7. (See Table 2.)

4. SET security hierarchical control model

4.1. Principle of safety hierarchical control model

The principle of the model is to divide the protocol into different security levels according to the actual needs of customers. The lower the level, the lower the security, but the fewer the number of identity authentication and encryption and decryption signatures, the faster the completion of the entire transaction; otherwise, The higher the level, the stronger the security, the greater the number of identity authentication, encryption and decryption signatures, and the longer it takes to complete a transaction. Customers can choose a security level that suits their actual needs based on their actual conditions. If the transaction amount of the product purchased by the consumer is small and the security requirements are not so high, in order to increase the transaction speed, you can choose a lower security method; if the transaction amount of the product purchased by the consumer is large, in order to ensure that For transaction security, you can choose a higher security level. Because the main protocols used in e-commerce are the SSL protocol and the SET protocol, and SET is more secure than SSL, therefore, we divide SSL and SET as the boundary: the security that is worse than SSL is classified as A; It is classified as Class B; the security is higher than SSL but lower than SET is classified as C; the unimproved SET protocol is classified as C; the improved SET protocol is classified as E. The improved SET protocol is mainly applicable to transactions with very large transaction amounts and extremely high security requirements. But the operation is more complicated than Class D. The security classification control is as follows:

(1) Customers, merchants, and payment gateways need to verify each other's certificates to verify the other party's identity is true and valid.

Table 2
The number of participants authentication.

Protocol	Customer	Merchant	Payment institution	Fourth party
SET protocol	2	2	2	
Secure e-commerce protocol based on the fourth party	1	0	3	0

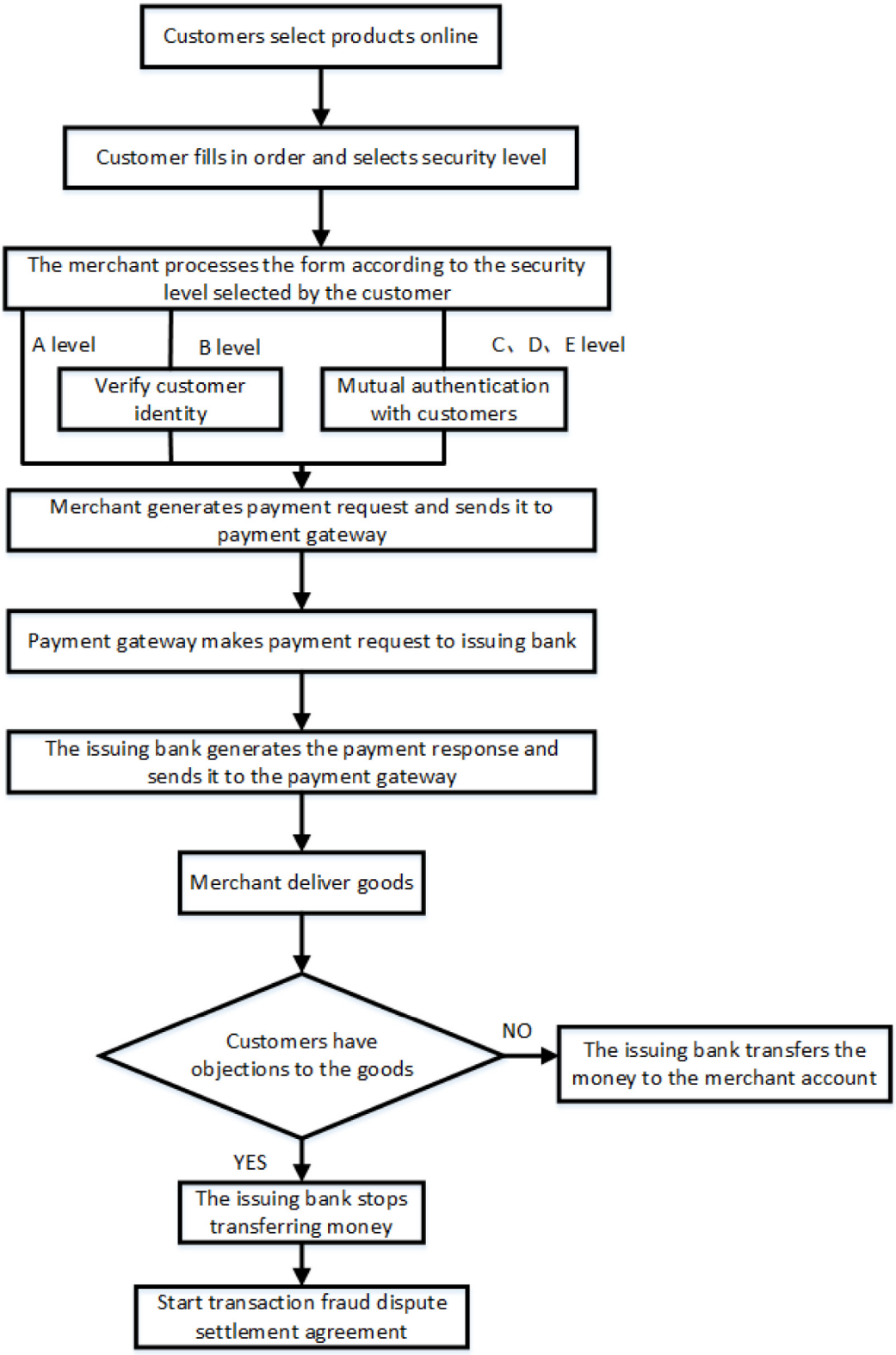


Fig. 8. Process of security control classification model.

(2) When encrypting the information to be transmitted, first use the algorithm developed by the country itself and proved to be mature and effective to re-encrypt, and then use the original encryption method of SET to perform secondary encryption to achieve nested encryption and enhance the delivery security of information.

(3) In order to be able to judge in the event of a dispute between the customer and the merchant, the issuing bank has the power to arbitrate. In this way, when the dispute occurs, the card issuer does not directly transfer the consumer's payment into the merchant's account when the transaction proceeds to the payment stage, but deposits the prepayment into a temporary account for three to seven days, until the consumer receives the correct product and no opinion, the payment in the temporary account is transferred to the merchant's account.

4.2. Implementation of the safety hierarchical control model

The flowchart of the safety hierarchical control model is shown in Fig. 8:

The specific implementation steps are as follows:

① Customers browse the online store to purchase the products they need, and fill in the order.

② The customer selects a security level according to actual needs as the later standard.

③ The customer sends the required security control level, order instruction and payment instruction to the merchant.

④ After receiving the instruction, the merchant generates a payment request instruction according to the security level specified by the customer and sends it to the payment gateway.

⑤ After receiving the instruction, the payment gateway selects the corresponding security level to request processing and sends the processing request payment instruction to the acquiring bank.

⑥ After the acquiring bank receives the request, it sends the request to the card issuing bank; the card issuing bank processes the payment request according to the corresponding security level. If it is Class E, the card-issuing bank does not immediately transfer the cardholder's payment to the acquiring bank, but temporarily transfers the payment to a temporary account to see if a dispute occurs and then resolves it.

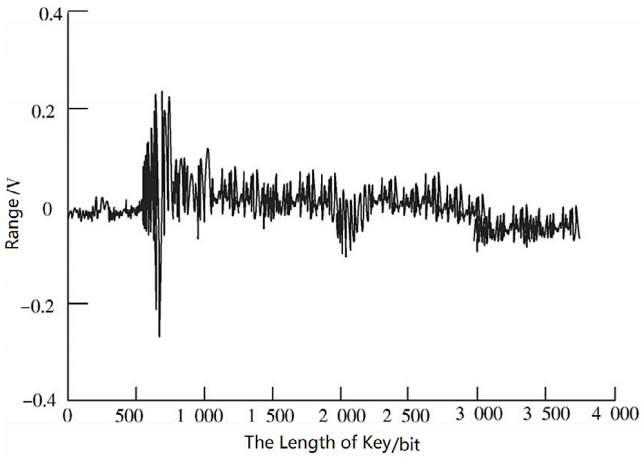
5. Evaluation of safety hierarchical control model

The security hierarchical control model is based on the actual situation. According to the size of the transaction amount and the customer's needs for transaction efficiency and security, different security levels are selected to better meet the needs of different consumer groups. So that customers have more initiative to the security they need, and it is also beneficial to customers who do not apply for certificates. It solves the disadvantages of set protocol, such as slow speed and complex operation, which are caused by the same processing method regardless of the actual needs of customers.

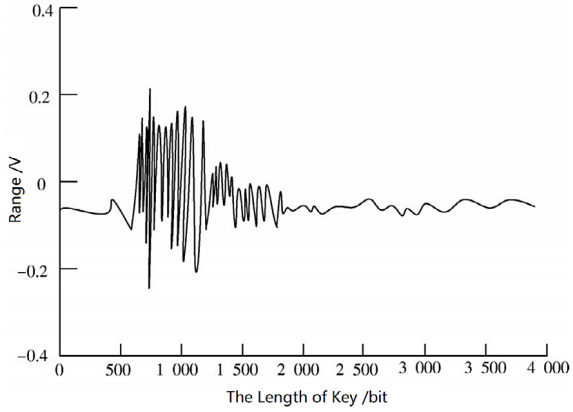
In order to verify the effectiveness of the improved model of secure e-commerce payment based on set protocol, a simulation experiment is designed.

Under different key length, the encryption, decryption and the number of key pair generation of this model and traditional payment model (traditional model) based on interactive multi biometric recognition technology are compared. The execution efficiency results of different models are shown in Table 3.

According to Table 3, we know that the encryption and decryption speed of the model encryption algorithm in this paper is 7–11 times faster than that of the traditional model encryption algorithm, under the same payment environment. It can be seen that the performance of the encryption algorithm of this model is close to the theoretical value, which can provide a high-speed, secure and stable e-commerce payment environment. The experiment compares and analyzes the e-commerce payment security provided by this model and the traditional model, and the results are shown in Fig. 9.



a) Traditional payment model payment security test curve



b) Payment security test curve of this payment model in this paper

Fig. 9. Security test results for secure E-commerce payments of two models.

By analyzing Fig. 9, we can see that the traditional model has a strong influence on the change of e-commerce payment security, and the curve is relatively rough; the model has a small change in curve and a high degree of smoothness, indicating that the model in this article can optimize and encrypt e-commerce payment. The security performance of the payment process has been greatly improved, so that e-commerce payments can be securely traded even in a poor channel environment. Experiments verify the effectiveness of the model in practical applications. E-commerce activities are conducted on different network platforms and the time required to complete an e-commerce payment is tested.

6. Conclusions

This article is to deal with the lack of security, low efficiency, and lack of a reasonable transaction dispute handling rule brought by SET's original protocol to formulate corresponding improvement measures. Aiming at the problems of low protocol security and weak scalability, a server e-wallet model was proposed, which divided the original "fat" wallet into an e-wallet client and an e-wallet server. The e-wallet client is used to receive the customer's input instructions during the communication process and send it to the e-wallet server. After the server-side processing is completed, the processing result is sent to the customer. It returns the processing result to the client. Because the wallet verification function is transferred from the client to the card-issuing bank or other financial institution and the client only uses a

Table 3
Comparison results for execution efficiencies of models.

Key Length	Encryption		Decryption		Number of Key	
	Traditional model	Current model	Traditional model	Current model	Traditional model	Current model
1024/160	7	10	6	17	20	244
3072/256	2	32	3	626	35	4126
15360/512	4	79	72	74048	76	858906

browser plug-in with very little memory, it is more convenient and more secure than the traditional e-wallet client. Aiming at the problem that the agreement cannot guarantee the atomicity of the goods and confirm the atomicity of sending, which will easily lead to disputes between consumers and merchants, a fourth-party secure e-commerce transaction protocol is proposed. In this agreement, a fourth party is added, which can be a logistics company or other authoritative express company. Its role is to supervise the timeliness and effectiveness of the information flow of the entire transaction between the parties to the transaction and to provide the results of the processing of disputes when necessary. The agreement is divided into four parts: the main agreement; the return agreement where the customer requires a return: the exchange agreement where the customer requires a replacement; the dispute settlement agreement where the product received by the customer is inconsistent with the customer's order. Compared with the original protocol, the improved protocol has only 4 times of identity authentication, and the original protocol needs to be verified 6 times. Therefore, the efficiency of the improved protocol authentication has been greatly improved.

The simulation experiments of the protocol model show that the improved protocol satisfies the requirements of commodity atomicity and confirmation sending atomicity, and can effectively deal with various situations. Aiming at the problems of complex and inefficient protocol transactions, a security control hierarchical model was proposed. In this model, according to the security level required by the customer and the size of the transaction amount, the transaction agreement is divided into 5 levels, namely A level, B level, C level, D level, and E level. According to the order of security level: A < SSL = B < C < SET = D < E, and once formulated the security standard control situation table at all levels, and gives the model implementation flowchart. The improved protocol selects different security levels according to the size of the transaction amount and the customer's needs for transaction efficiency and security, which better meets the needs of different consumer groups, and better solves the slow speed and complex operation of the SET protocol. And other disadvantages, met the expected requirements.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Bing Xu: Funding acquisition. **Darong Huang:** Writing - original draft. **Bo Mi:** Conceptualization.

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